ENGINE ARCHITECTURE REFERENCE

SPEC-Engine Architecture Reference

SCOUT SPEC v3.1

Complete Signal Catalogs, Scoring Logic & Cross-Engine Comparison

All Pools, Timeframe Weights, Thresholds & Helper Functions

Engine 1: SCOUT SPEC (v3.1)

The SCOUT SPEC engine is the primary entry-scoring system designed for early detection of recovery opportunities following RSI troughs. It uses a multi-pool, multi-timeframe scoring architecture with 25 signals across 5 pools, producing a total score that drives trade entry decisions. Version 3.1 introduced gradual scoring across all Pool B signals to eliminate binary flip-flop behavior.

1.1 Scoring Formula

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Total Score = min(90, Pool_A + Pool_B + Pool_C) + Bonus_A + Bonus_B
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Component	Max Points	Signals	Scoring Nature
Pool A (RSI Recovery)	45	10	Gradual power-curve + spread-based
Pool B (Price Structure)	30	5	Gradual / distance-based (v3 redesign)
Pool C (Confirmation)	12	3	Binary fire-and-hold + gated
Bonus A (Structural)	12	3	Binary stepped / composite
Bonus B (Uptrend Correction)	13	5	Gated by BB1 (uptrend confirmed)
Core Cap	90	20 signals	min(90, A+B+C)
Grand Maximum	115	25 signals	90 + 12 + 13

Table 1: SCOUT SPEC scoring structure

1.2 Timeframe Architecture

The engine operates on three timeframes (4H, Daily, Weekly) with explicit per-signal weights. Weekly carries the highest weight for most Pool A signals, reflecting its importance in confirming trend reversals. Pool B uses 4H and Daily for convergence/spread signals and all three TFs for proximity. The WAP21 signal (B5) is Daily-only.

1.3 Pool A: RSI Recovery (45 pts, 10 signals)

Pool A forms the backbone of the SCOUT scoring system, measuring how effectively RSI indicators are recovering from trough conditions. Eight of the ten signals use gradual scoring that ramps smoothly as recovery progresses, eliminating the binary on/off behavior that plagued earlier versions.

A1: RSI21 Reversal (max 7.5 pts, Gradual Power-Curve)

Measures recovery from the cycle-low RSI21 using a power-curve formula: recovery_ratio = (current_RSI21 - cycle_low) / (100 - cycle_low), then points = max_pts * min(1.0, recovery_ratio ^ 0.7). The 0.7 exponent means early recovery earns points faster than linear, rewarding the initial bounce more heavily. Applied across all three timeframes with weights: 4H=1.5, D=2.5, W=3.5.

A2: RSI36 Reversal (max 3.0 pts, Gradual Power-Curve)

Identical power-curve logic applied to RSI36. TF weights: 4H=0.5, D=1.0, W=1.5. Captures medium-term RSI recovery.

A3: RSI56 Reversal (max 2.0 pts, Gradual Power-Curve)

Same power-curve on RSI56, the slowest RSI period. TF weights: 4H=0.5, D=0.5, W=1.0.

A4: Multi-Speed RSI Divergence (max 3.5 pts, Fire-and-Hold)

Uses a 30-bar lookback split into two 15-bar halves. Finds the local price minimum in each half. Requires the recent price low to be lower than the prior price low (price lower-low). Then checks RSI21, RSI36, and RSI56 at those two price lows for bullish divergence (RSI making a higher-low while price makes a lower-low). Requires at least 2 RSI periods showing divergence to score (2 periods = 60% weight, 3 periods = 100%). Fire-and-hold: once triggered, persists until next trough. TF weights: 4H=0.5, D=1.0, W=2.0.

A5: RSI21 > RSI36 Crossover Spread (max 7.5 pts, Gradual)

Measures HOW FAR RSI21 is above RSI36, not just whether it is above. Formula: scale = clamp((RSI21 - RSI36) / 4.0, 0, 1). A 4-point spread range was calibrated from the 75th percentile of observed spreads. Linear ramp from 0 spread to 4+ spread. TF weights: 4H=1.5, D=2.5, W=3.5.

A6: RSI21 > RSI56 Crossover Spread (max 5.0 pts, Gradual)

Same spread logic for the wider RSI21-RSI56 pair: scale = clamp((RSI21 - RSI56) / 7.0, 0, 1). The 7.0 range reflects the naturally wider spread between fast and slow RSI. TF weights: 4H=1.0, D=1.5, W=2.5.

A7: RSI36 > RSI56 Crossover Spread (max 3.0 pts, Gradual)

Spread logic for the medium-slow pair: scale = clamp((RSI36 - RSI56) / 3.0, 0, 1). Tightest range at 3.0 points since RSI36 and RSI56 track more closely. TF weights: 4H=0.5, D=1.0, W=1.5.

A8: RSI Recovery Momentum (max 7.0 pts, Gradual)

Measures the 3-bar rate of change in RSI21: momentum = (RSI21_current - RSI21_3bars_ago) / 3. Only scores if momentum exceeds 0.5 (positive recovery). Normalized: scale = clamp((momentum - 0.5) / 2.0, 0, 1). TF weights: 4H=1.5, D=2.5, W=3.0.

A9: Cross-TF Recovery Alignment (max 4.0 pts, Binary Stepped)

Counts how many timeframes simultaneously show positive A8 momentum. 2 TFs = 2.0 pts; 3 TFs = 4.0 pts; 0-1 TFs = 0. Rewards multi-timeframe alignment of recovery momentum.

A10: Multi-TF Trough Agreement (max 3.0 pts, Binary Stepped)

Counts how many timeframes are currently in an active recovery window. 2 TFs = 1.5 pts; 3 TFs = 2.5 pts. Confirms that the trough was a multi-timeframe event, not a single-TF blip.

1.4 Pool B: Price Structure (30 pts, 5 signals) - v3 Gradual Redesign

Pool B was completely redesigned in v3 to eliminate binary scoring. All signals now use distance-based linear ramps, ensuring scores change gradually as price structure evolves. No signal flips from 0 to maximum on a single bar.

B1: TP28 Convergence (max 3.5 pts, Gradual, 4H+Daily)

Uses _convergence_score(): measures how tightly bunched the TP28 (Typical Price 28-bar moving average) values are over the last 10 bars. Coefficient of variation (CV) is calculated as (max - min) / |mean| x 100. Linear ramp: CV = 0% = full points; CV >= 15% = 0 points. Requires at least 6 non-NaN values. TF weights: 4H=2.0, D=1.5.

B2: WC50 Convergence (max 3.5 pts, Gradual, 4H+Daily)

Same convergence logic applied to WC50 (Wilder's Close 50-bar moving average). Flat WC50 indicates reduced volatility and price consolidation, a prerequisite for directional moves. TF weights: 4H=2.0, D=1.5.

B3: Price to TP28 Proximity (max 9.5 pts, Gradual, All TFs)

Uses _proximity_score(): linear ramp measuring how close price is to TP28. If price is at or above TP28: full points immediately. If below: dist_pct = (TP28 - Close) / TP28 x 100; scale = 1.0 - (dist_pct / 12.0). At 12% below TP28, score reaches zero. This provides the strongest single signal in Pool B. TF weights: 4H=2.5, D=3.5, W=3.5.

B4: TP28-WC50 Spread (max 6.0 pts, Gradual, All TFs)

Uses _spread_score(): measures how much TP28 has pulled above WC50. spread_pct = (TP28 - WC50) / |WC50| x 100. Linear ramp: 0% spread = 0 points; 10%+ spread = full points. A positive and growing spread indicates that the shorter-term average is leading the longer-term average upward. TF weights: 4H=1.5, D=2.5, W=2.0.

B5: Price vs WAP21 (max 7.5 pts, Gradual, Daily-Only)

Uses _price_wap_score(): computes a 21-bar linearly weighted average of close prices (recent bars weighted more heavily). Two-component scoring: Distance component (60% weight) measures how far price is above WAP, with a 5% above-WAP ceiling for full distance points and a near-zone (-3% to 0%) providing partial credit. Persistence component (40% weight) counts consecutive bars where Close > WAP x 0.98 (2% tolerance), scaling linearly over 21 bars. This signal directly addresses the need for price-based weightage in the scoring system.

1.5 Pool C: Confirmation (12 pts, 3 signals)

Pool C provides the final confirmation layer. These signals are mostly binary, acting as gates that confirm the recovery is genuine before a trade entry is considered.

Signal	Max Pts	Type	Logic
C1: Weekly Divergence	5.0	Binary fire-and-hold	Weekly-only, 20-bar lookback, price LL + RSI21 HL
C2: D-RSI21 > 50	3.0	Binary live/die	Daily RSI21 above 50
C3: Cross-TF Boost	4.0	Binary gated	Requires >=2 TFs with crossovers active AND B3 score > 0

Table 2: SCOUT SPEC Pool C confirmation signals

1.6 Bonus Pools

Signal	Max Pts	Type	Logic
BA1: Volume Surge	5.0	Binary stepped	Vol/AvgVol: >1.5x=2, >2.5x=4, >3.5x=5
BA2: Support Bounce	3.0	Binary stepped	>=15% decline from 180-bar peak; proximity to 85% level
BA3: Trend Quality	4.0	Composite	MA structure + rising lows + correction compression + RS proxy
BB1: Uptrend Confirmed	2.0	Binary gate	MA200 rising AND Close > MA200 (prerequisite for BB2-BB5)
BB2: Correction Depth	5.0	Binary stepped	From 60-bar high: 25-35%=3, 35-45%=4, 45%+=5
BB3: RSI Turning Positive	2.5	Per-TF live/die	Any RSI crossing above 50; 4H=0.5, D=1.0, W=1.0
BB4: Price Above Support	2.0	Binary	Close > 85% of 180-bar structural peak
BB5: Volume on Recovery	1.5	Binary	>=3 up-days with avg up-day vol >= 1.2x overall avg

Table 3: SCOUT SPEC bonus pool signals

1.7 Classification & Entry Thresholds

Score Range	Classification	Action
0 - 49	No Entry	No action
50 - 64	Pilot Buy	Small position, 25-30% sizing
65 - 79	Standard Entry	Standard position
80 - 89	Full Entry	Full position
90+	Full Strong	Maximum position (rare)

Table 4: SCOUT SPEC classification tiers

Thresholds are calibrated for the gradual scoring system and are equivalent in quality to the old binary 70 threshold. The exit trigger fires when the score drops below 50 while in position. No stop-loss, trailing stop, or time-based exit is implemented; the system relies entirely on score degradation for exit signals.

1.8 Trough Detection & Recovery Window

Trough detection uses detect_troughs(series, rsi_col='RSI21', lookback=11): RSI21 must be <= 40 and be the minimum within an 11-bar centered window. Applied independently on all 3 timeframes. After each trough, the next 60 bars (RECOVERY_WINDOW = 60, reverted from 100) are marked as in_recovery = True. Only bars within the recovery window receive non-zero scores.